Simulation of Base Ahmed body and its Validation with Experimental data [Case 8]

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Abstract

In this study, a two-dimensional (2D) Ahmed body is numerically simulated using computational fluid dynamics in Star CCM+. Initially, a 25° slant angle was envisaged for the 2-D model. Then, both models are subjected to simulations of unstable flow with the slant angle changed to 12.5°. Analysis is done on how the change in slant angle affects the variation in the drag and lift coefficients. Evaluating how our results differ from Lienhart and Becker's experimental findings.

Keywords: Ahmed body, drag coefficient, lift coefficient, Slant angle

1. Introduction

Reverse flow characteristics and large-scale separations on the vehicle serve as indicators of the turbulent nature of the flow around any given object. Understanding the flow dynamics around the vehicle is vital before creating any new models of it. When employing computational fluid dynamics (CFD) both computationally and experimentally, the Ahmed body is regarded as the fundamental problem for the evaluation of flow dynamics for all four-wheeled vehicles. The automotive industry constantly seeks to lower drag force to boost fuel efficiency. Aerodynamic drag and lift force are key characteristics to analyse the performance of the vehicle. According to the research, a 10% reduction in drag force results in a 7% reduction in fuel consumption.

The three-dimensional simplified model of the vehicle was created by Ahmed et al. to help comprehend flow dynamics because real-world vehicles are difficult to represent numerically and experimentally. Ahmed body should ideally have a length of 1044mm, a height of 288mm, and a constant slant length of 222mm for any angle in the rear. The slant portion of the vehicle have an impact on aerodynamic drag.

Static pressure measurements were made by Hermann Lienhart and Stefan Becker to examine the flow and turbulence structure surrounding a simplified car model. To provide reference data set for numerical simulations in general and the verification and validation of improved turbulence models in particular, a detailed data collection was collected under clearly defined boundary conditions. The study concentrated on the wake behind a simple car model due to the losses in the separated wake region making the largest contribution to the aerodynamic drag and the accuracy rate of the wake being a very selective parameter for the turbulence models.

2. Base Ahmed Body

The software we used to create the 2D Ahmed body model was Star CCM+. Based on the issue presented in case 8, all measurements are in millimetres, also they are depicted in metres in the figures.

2.1 Geometric description of base Ahmed body and computational domain

Table 1 below lists all the geometric dimensions of the Ahmed body and the fluid domain that surrounds it. Initial numerical simulations are conducted using a 25° slant angle. Figure 1a depicts the two-dimensional Ahmed body model for a slant angle of ϕ = 25°. Figure 1b shows the dimensions of computational domain with respect to Ahmed body. The domain is set at a distance of 20.5 times the projected length from the rear end of Ahmed body. At 1328 mm upstream from the front face of the Ahmed body, the flow condition is evaluated.

Table 1. Dimensions of base Ahmed body and computational domain

Height	288mm
Length	1044mm
Ground clearance	50mm
Slant angle (ϕ)	25°
Front radius	100mm
Fluid domain	8276mm x 1382mm

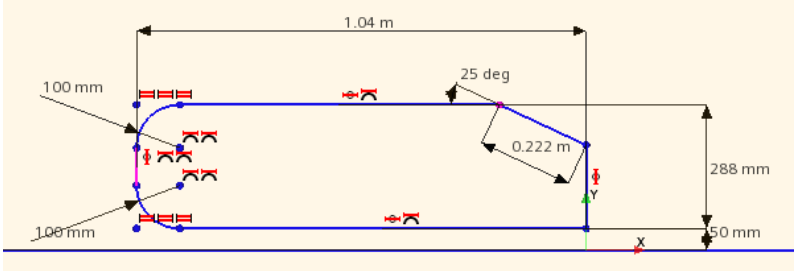


Figure 1a. 2D drawing of Ahmed body with its geometric dimensions

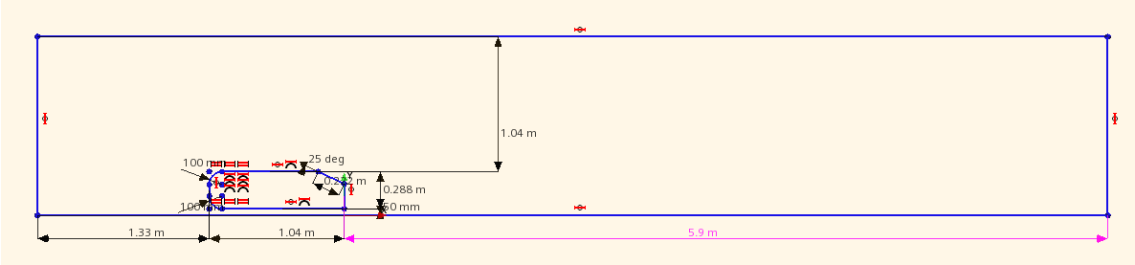


Figure 1b. Dimension of Computational Domain with respect to Ahmed body

Table 2 lists all the calculated values for various dimensions for a 25° slant angle, and Figure 2 displays the corresponding model. In the table below L represents the horizontal distance covered by slant and H represents the vertical distance covered by slant.

Table 2. Calculated values for 25° slant angles

Slant angle (ϕ)	25°
X (mm)	843mm
Z (mm)	194mm
L(mm)	201mm
H (mm)	94mm

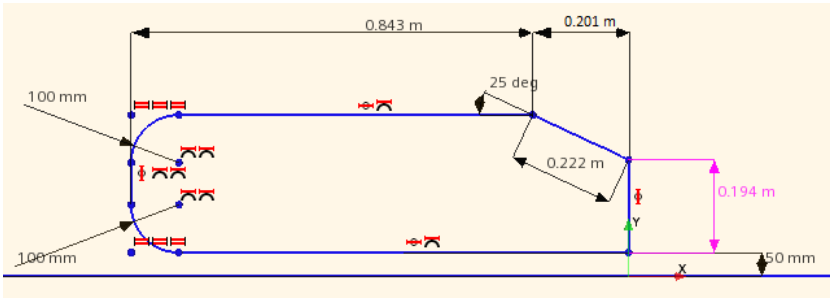


Figure 2. Calculated values supporting table 2

2.2 Meshing

'Trimmer' is the type of mesh that is chosen to produce hexahedral mesh. The 'Prism layer Mesher' is turned on to enable the creation of layered mesh close to wall boundaries. The software would attempt to mesh the computational domain based on the base size, therefore a 5mm base size and 'Number of Prism layer' as 10 were chosen to ensure that computation time is reasonable while yet attempting to get an accurate result.

The 'Relative Minimum Size' is set to 25% and the 'Relative Maximum Size' to 100% in order to minimize the number of cells in the area of focus. To allow for a gradual change in cell size, the Slow 'Default Growth Rate' and 'Boundary Growth Rate' options are set. Prism layer mesh was disabled for all other boundaries except for Ahmed body, to limit the created prism layer mesh to the Ahmed body boundary.

2.3 Numerical Modelling

Under 'Physics continuum' Implicit unsteady is chosen to find effects of unsteady flow simulation, Gas is selected by assuming that the flow situation requires air, Segregated flow is selected to allow the solver to perform each equation in sequence, by considering the flow to be incompressible Constant Density is selected, and 'K-omega Turbulence' is selected because it is one of the Reynolds-Averaged Navier-Stokes (RANS) turbulence models that is appropriate for the flow having separation. The programme automatically engages All y+ Wall Treatment when K-Omega Turbulence is enabled to treat the boundary layer next to the wall boundaries. Following this, the mesh is generated and converted to 2D with the intention of lowering the computational cost, Figure 3 illustrates the mesh that was obtained.

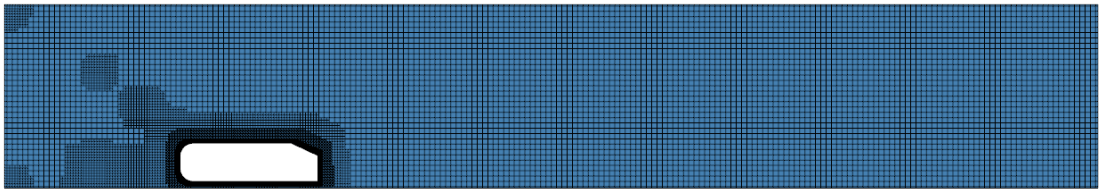


Figure 3. 2D Mesh of domain

2.4 Mesh Refinement

Refining the mesh is necessary to increase the solution's accuracy. It is possible to refine the mesh for the entire domain but doing so would be time consuming and would not be cost efficient. As a result, we could limit our attention to the area surrounding Ahmed's body surface, where the flow is most important. Also, the flow around downstream of the Ahmed body should also be refined appropriately.

Within the computational domain, a small box with a fixed mesh size is constructed around the Ahmed body. The box is stretched all the way to the limit of the outlet. Prism layer thickness been increased to 50% and the number of layers has been increased to 20. For the box to be meshed by the 5mm hexahedral mesh, the proportion of base mesh at this point is set to 100%. A new mesh is generated as shown in Figure 4, after the previously generated mesh has been cleared. Following the refinement, 35135020 verts, 100468060 faces, and 33624276 cells are generated. When the 3D model is converted to 2D, there are 428629 cells, 856697 faces, and 435444 verts.

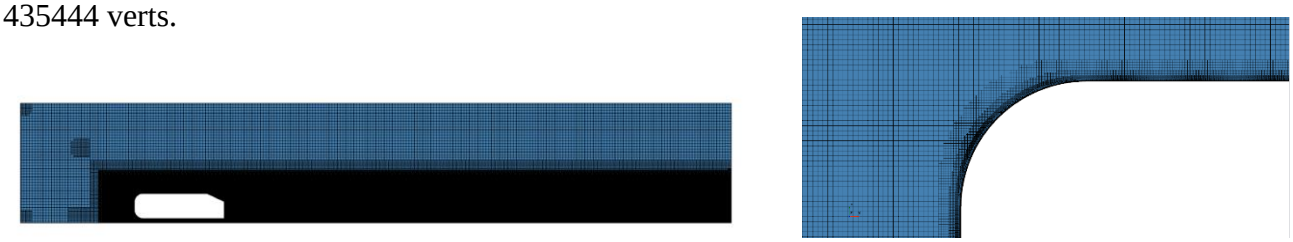


Figure 4. Refined mesh

2.5 Boundary Conditions

The flow conditions were set based on Lienhart and Becker's experimental work. At the location of the measurement, the turbulence intensity is set to 0.0025 and the average approaching flow stream velocity is set to 40m/s. Bottom wall boundary is set under No slip condition to ensure that the velocity at the wall is zero. The top wall boundary is modified to slip to allow flow to move along the boundary.

2.6 Solver setup for Numerical Simulation

This setup is a must to terminate the simulation. The maximum number of iterations is typically one of the options for stopping the simulation. However, a big number should be provided, and we have provided a maximum step of 5000. Continuity, X momentum, Y momentum, the specific dissipation rate, turbulence kinetic energy was used to ensure that the simulation would end only when it reached a particular degree of accuracy. During the simulation these are tracked and are set to a minimum limit of 1.0E-4. The logical rule is modified from ‘Or’ to ‘And’ in order to ensure that these requirements are accomplished at all costs.

Drag and lift coefficient graphs are displayed to understand how the flow over Ahmed's body affects how the two coefficients behave over time. The reference area is chosen as 0.288 m^2 because the projected area along the flow over a 2D Ahmed body is equal to the height of the

Ahmed body, and the reference density is given as 1.2 kg/m^3 because air is the fluid being considered. These values are all set in the drag coefficient property window. By simply shifting the direction to [0.0, 1.0, 0.0] as the lift force acts vertically on a body, the same parameters are followed for the lift coefficient. For implicit unsteady, the time step is set to 0.001s. By initializing and running, the simulation starts and would end when the predetermined criteria are met.

2.7 Results and Validation

Figure 5 displays the residual monitor plotted for the unsteady simulation. The solvers' proper behaviour, which is indicated by generally smooth fluctuation with increasing iterations, can be observed using the residual monitor for unstable simulation. The profiles and actual residual values themselves are not significant, it is seen that continuity and Y-momentum are considerably high as per the graph. Maximum inner iteration is set to 5.

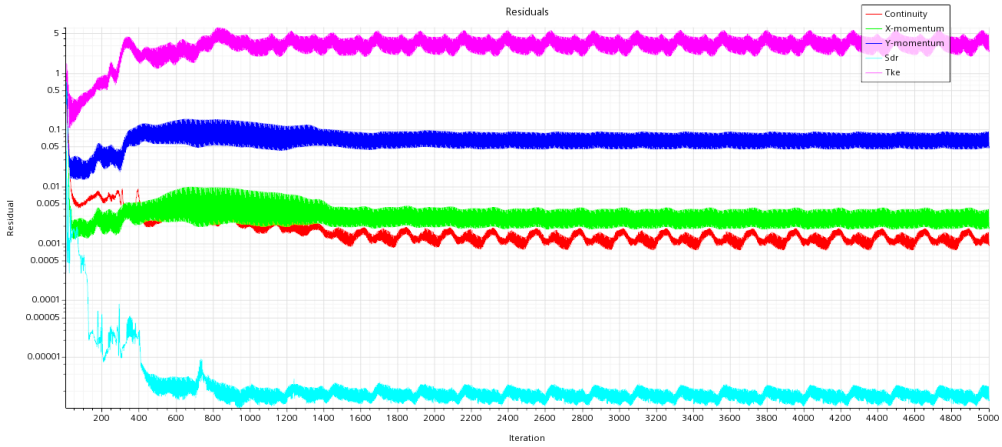


Figure 5. Unsteady Residuals monitor

Figure 6 displays the Ahmed body's velocity streamlines, which illustrate the velocity change brought on by a 25° slant angle. The downstream flow separation and the recirculation zone close to the slant angle are responsible for the majority of the drag on Ahmed's body. It is necessary to sample the data over a specific time period in order to acquire the mean data. The mean velocity in X direction over Ahmed's body is shown in Figure 7.

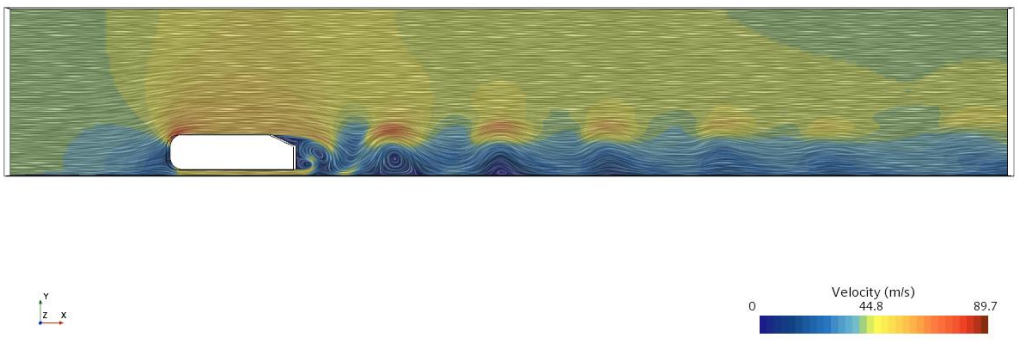


Figure 6. Velocity vector for slant angle ϕ =25°



Figure 7. Scalar Mean velocity in X direction

The pressure contour for Ahmed body's 25° slant angle is shown in Figure 8. It is evident from the figure that the wake region and the curved front part of the vehicle experience the highest pressure.

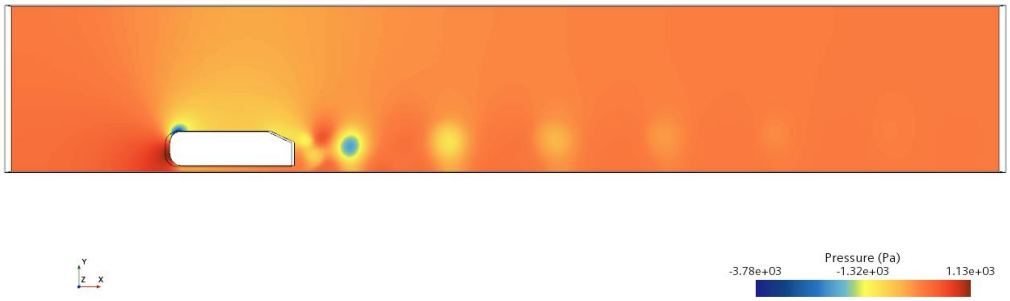


Figure 8. Pressure contour for 25° slant angle

As demonstrated in Figure 9, the drag coefficient and lift coefficient along physical time for a slant angle of 25° are combined into a single plot. Maximum physical time is set to 1.0s to attain the stable stage where the drag and the lift coefficients oscillate periodically at the same amplitudes. C_D and C_L curves initially exhibit fluctuating values due to flow instabilities, but as physical time rises, the values become stable. After 0.35s of physical time, these figures demonstrate constant oscillation. Thus, the average Drag and lift coefficient values are derived from the point at which the plot becomes stable. Tabulating the results yielded a drag coefficient of 0.58 and a lift coefficient of 2.15.

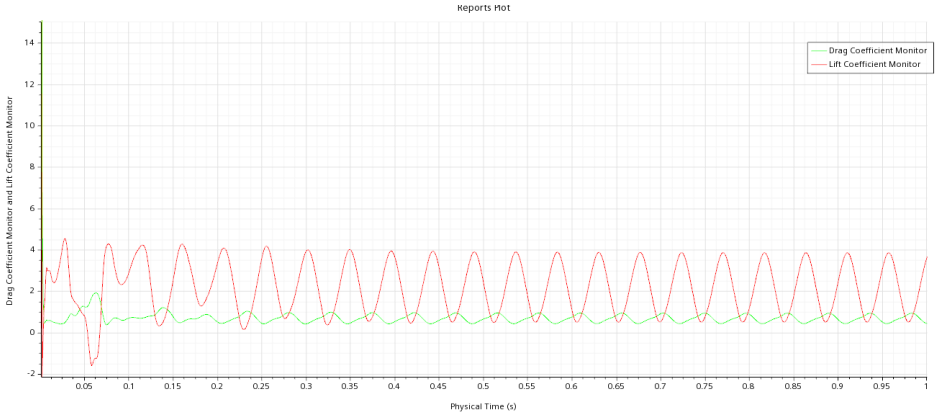


Figure 9. Drag and Lift Coefficient along Physical time

In the experimental work, Lienhart and Becker claim that the Ahmed body was chosen for testing because of its relatively simple geometry, which made it easier to generate the grid for the numerical predictions but still demonstrated the complexity of the flow field surrounding a car-like shape. The experiment was mainly concerned with the wake left by a simplified car. Figure 10 depicts a graph of drag coefficients for various slant angles together with a budget of individual contributions taken from the experimental work of Lienhart and Becker. In the figure C_w is the coefficient of Total drag.

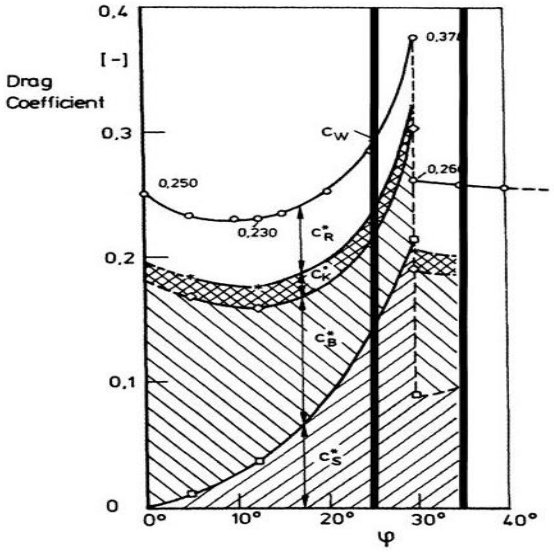


Figure 10. Drag coefficient for varying slant angles

Lienhart and Becker conducted their experiment on flow over a 3D Ahmed body, whereas we have converted the 3D model to 2D and performed simulations for flow over a 2D Ahmed body, this is the reason for the discrepancy between the drag coefficient values that are obtained through our simulation and the experimental data.

3. Design Proposal

The basic model is altered to evaluate the impact of flow caused by a change in geometry.

3.1 Proposed New Design

We are required to reduce the slanted angle to 12.5° in accordance with the provided case. Figure 11a and 11b represents the geometry of the proposed design.

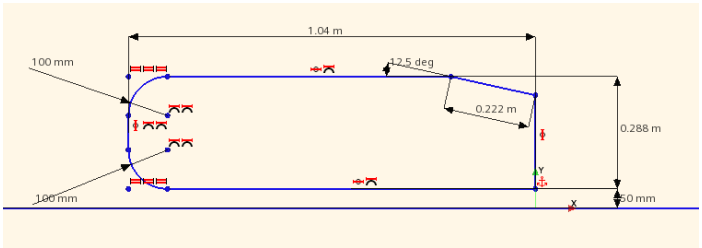


Figure 11a. Ahmed body geometric dimensions with 12.5° slant angle

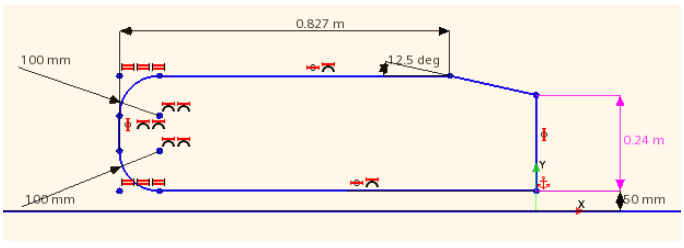


Figure 11b. Dimensions of proposed design with 12.5° slant angle

3.2 Results and comparison

Boundary conditions and other Numerical simulation setups are similar to that of the base model. The results obtained after simulating the Flow over Ahmed body with the ϕ =12.5° are as follows. If we carefully observe the flow movement between Figure 12 and Figure 6. It is evident that there is flow separation closer to slant angle in case of ϕ = 25°, whereas in case of ϕ =12.5°, the flow is continuous over the slant angle. Additionally, we can also see that there is less vortex in Figure 12 when compared with Figure 6.

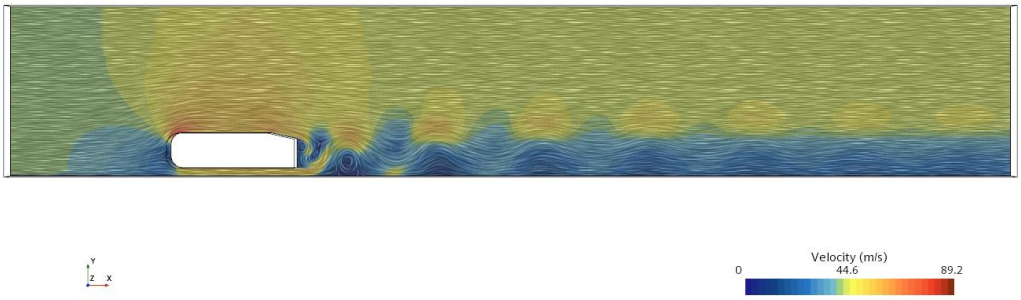


Figure 12. Vector scene for 12.5° slant angle

Figures 13 and 14 depict the scalar scene of mean flow velocity in the X direction at 12.5° slant angles and pressure contour respectively. In Figure 13 the wake is created at the point where slant angle ends and that's where the flow separation occurs. Alos the flow downstream is as not smooth as 25° slant angle. According to Figure 14, the curved front portion has the highest pressure. When we compare this with Figure 8, we can see that pressure is being applied at the point where slanted angle ends which is not the case in Figure 8.

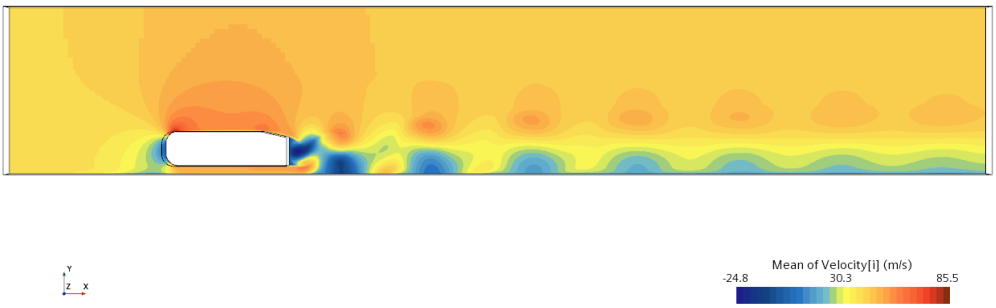


Figure 13. Scalar scene of mean velocity for ϕ =12.5°



Figure 14. Pressure contour for ϕ =12.5°

Figure 15 below shows the plot for residuals over iterations, and Figure 16 shows the drag and lift coefficient along physical time for a new proposed design's unstable flow. We obtain drag coefficient of 0.54 and lift coefficient of 2.6 if the average of drag and lift is calculated from the location of steady periodic oscillations. Drag coefficient has decreased and lift coefficient has increased when these numbers are compared to the initial Ahmed body. The drag coefficient in from Lienhart and Becker is approximately 0.23 for 12.5°

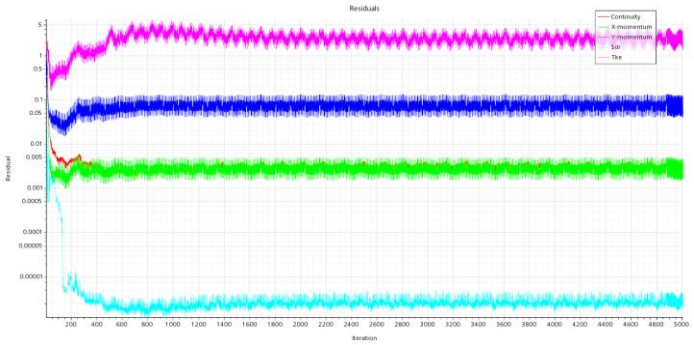


Figure 15. Residuals for over iteration ϕ =12.5°

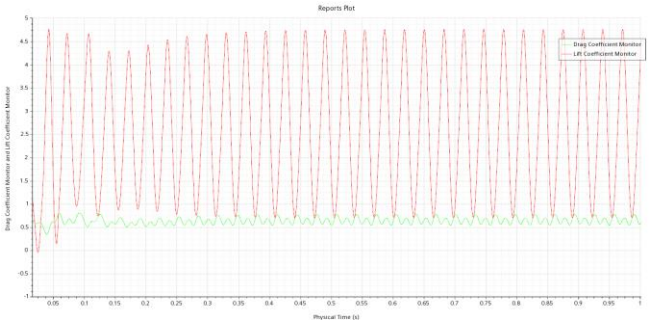


Figure 16. Drag and lift coefficient over physical time for ϕ =12.5°

4. Conclusion

In the current work, numerical computation of flow dynamics around 2D Ahmed body car model for base model with slant angle of 25° and the new proposed design with slant angle of 12.5° is created. The drag coefficient decreased with reduction in slant angle while the lift coefficient increased. The increase in lift for condition of ϕ =12.5° is because the large vortex is formed closer to the rear bumper. The experimental data was obtained by simulating a 3D Ahmed body mounted over cylindrical stilts, but our numerical simulation was done for a 2D Ahmed body, which is why there were disparities between the values from the experimental data and our simulation. Mesh size, time step size, and modelling all affect the outcome of 2D simulations to get an accurate result, thorough simulation setup and modelling are needed. The mesh refining process can be improved further by extending the volume representation box's distance from the Ahmed body.